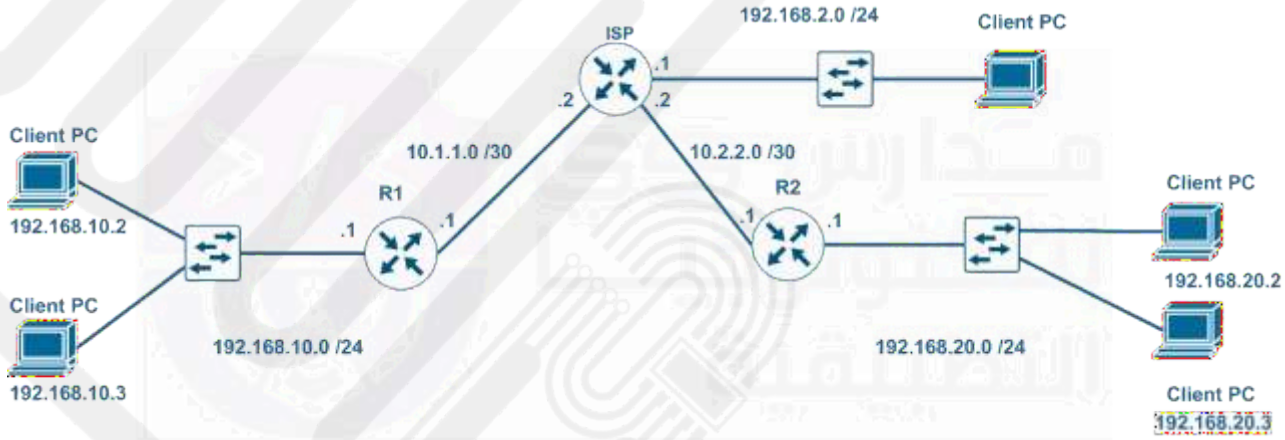
Exercise	**L**
Name	**IPsec configuration.**
Outcomes and Performance Criteria	➢ **Use packet tracer (CISCO simulation tool) and GNS3 emulation.** ➢ **Maintain network security.** ➢ **Secure organizations from threats**
Skills	TPC4.1: Identify, install, and configure different network infrastructure topologies TPC4.2: Path Metric calculations for different routing protocols TPC4.3: Configure devices to communicate on a network and virtual private network TPC4.4: Perform troubleshooting procedures and analyze problems TPC5.1: Apply security measures on an infrastructure network
Time Required	**90 mins**
Tools needed	Cisco Packet Tracer



You are required to configure a Site-to-Site IPsec VPN between R1 and R2 to securely allow traffic between their respective LANs (192.168.10.0/24 and 192.168.20.0/24). The ISP router acts as a pass-through device and is not involved in VPN configuration. Additionally, you will secure R1 and R2 with basic device access settings, including passwords and SSH.

- **Task 1: Secure Device Access on R1 and R2**

Console and VTY Passwords:
Set the password for the console line as WE1.
Set the password for the vty lines as WE2.
Enable Password:
Configure the enable password as WEstd.
SSH Configuration:
Create a username admin with the password adminwe.
Enable SSH access for remote management.
Ensure that the vty lines allow only SSH access.

- **Task 2: Configure the IPsec VPN Tunnel**

Create IKE Phase 1 Configuration on R1 and R2:
Set the following parameters:
Encryption: AES 256
Hashing: SHA-256
DH Group: Group 5
Authentication Pre-shared Key (PSK): WE123

Create IKE Phase 2 Configuration on R1 and R2:
Set the following parameters:
Protocol: ESP
Encryption: AES 256
Hashing: SHA-256

Define the VPN Traffic:
Traffic between 192.168.10.0/24 (R1's LAN) and 192.168.20.0/24 (R2's LAN) should pass through the VPN tunnel.

Apply the IPsec Policy:
Ensure the policy is applied to the appropriate interfaces (R1's WAN and R2's WAN).

Verification

1. Verify the VPN tunnel with:
    - show crypto isakmp sa
    - show crypto ipsec sa
2. Test connectivity between **LANs**:
    - From **Client PC** in 192.168.10.0/24, ping a **Client PC** in 192.168.20.0/24.
    - Open a SSH session from Client PC 192.168.10.2 to R2.
    - Open a SSH session from Client PC 192.168.20.2 to R1.